

USAF ACADEMY AF, CO, USA

WMO: 745310

Lat: 38.967N

Lon: 104.817W

Elev: 6572

StdP: 11.53

Time Zone: -7.00 (NAM)

Period: 07-19

WBAN: 93065

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-1.5	3.7	-12.7	3.5	23.0	-8.3	4.5	24.5	35.8	47.8	29.0	34.7	7.8	0	0.637

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	28.1	89.3	56.2	86.8	56.6	84.4	56.3	62.4	75.4	61.2	74.6	60.2	73.8	11.3	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.3	96.4	64.2	57.6	90.7	63.6	56.1	85.9	63.9	31.4	75.6	30.4	73.8	29.5	74.1	66.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.0	23.9	20.2	DB	-11.7	93.6	5.7	1.9	-15.8	94.9	-19.2	96.1	-22.4	97.1	-26.5	98.5	(4)
(5)			WB	-12.4	65.7	5.6	1.0	-16.4	66.4	-19.7	67.0	-22.8	67.6	-26.8	68.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.5	29.9	30.3	39.0	44.2	52.3	64.4	68.8	66.6	60.6	47.0	37.8	28.0	(6)	
	DBStd	16.72	10.28	11.42	9.76	8.58	8.37	6.07	4.09	4.25	6.38	9.61	9.87	10.96	(7)	
	HDD50	3050	625	552	356	211	77	1	0	0	7	164	375	682	(8)	
	HDD65	6690	1090	970	805	624	397	79	15	32	156	559	817	1146	(9)	
	CDD50	2137	1	2	16	37	148	434	583	514	323	70	8	1	(10)	
	CDD65	301	0	0	0	0	2	63	133	81	22	0	0	0	(11)	
	CDH74	5586	0	0	1	8	181	1452	1941	1292	667	44	0	0	(12)	
	CDH80	1784	0	0	0	0	25	537	724	364	131	3	0	0	(13)	
(14) Wind	WSAvg	9.2	8.8	10.0	10.0	11.0	9.9	9.7	8.3	7.9	8.6	9.0	8.9	8.9	(14)	
(15) Precipitation	PrecAvg	20.6	0.6	0.6	1.3	2.5	2.5	2.0	3.5	3.4	1.6	1.1	0.7	0.6	(15)	
	PrecMax	30.1	1.7	1.7	3.7	8.3	9.9	5.9	6.5	5.9	5.8	2.7	2.9	2.0	(16)	
	PrecMin	11.0	0.1	0.1	0.3	0.2	0.6	0.2	1.2	0.6	0.0	0.1	0.1	0.2	(17)	
	PrecStd	4.4	0.3	0.4	0.8	1.7	2.0	1.4	1.5	1.3	1.2	0.7	0.6	0.4	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.8	65.3	72.2	75.1	83.2	93.6	91.7	89.3	87.1	78.4	70.0	61.5	(19)	
		MCWB	40.7	42.1	44.7	46.3	52.0	54.5	57.3	56.7	55.9	49.3	44.1	40.3	(20)	
	2%	DB	57.2	59.2	67.6	71.2	79.1	89.7	89.3	86.5	83.6	74.6	65.6	56.2	(21)	
		MCWB	38.4	39.0	42.6	45.7	49.7	54.4	57.2	57.3	53.9	48.1	42.6	37.8	(22)	
	5%	DB	53.1	55.4	63.5	67.9	75.2	86.2	87.0	84.2	81.0	71.2	61.5	52.3	(23)	
		MCWB	36.3	37.5	41.1	44.5	49.5	53.9	57.8	57.2	53.8	47.3	41.3	36.2	(24)	
	10%	DB	48.4	50.7	59.1	63.8	70.9	82.6	84.6	81.6	78.2	67.2	57.2	48.0	(25)	
		MCWB	34.1	35.1	39.3	43.2	49.1	53.9	58.0	57.3	53.3	46.3	39.2	33.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.1	42.7	45.5	50.4	56.1	62.3	64.4	64.0	60.8	54.8	45.5	41.0	(27)	
		MCDB	60.1	63.3	69.1	61.4	70.6	76.2	75.2	75.8	75.1	63.7	65.9	58.9	(28)	
	2%	WB	38.9	39.8	43.7	48.1	53.9	60.0	62.8	62.2	58.5	51.9	43.8	38.5	(29)	
		MCDB	55.3	57.4	65.2	63.2	68.6	74.6	75.9	74.6	72.0	66.2	63.9	54.8	(30)	
	5%	WB	36.6	37.8	41.7	46.2	52.3	58.5	61.7	60.8	57.1	49.4	41.9	36.4	(31)	
		MCDB	52.2	54.7	61.0	63.0	66.5	74.0	75.1	73.9	70.8	64.3	60.2	51.5	(32)	
	10%	WB	34.4	35.4	39.9	44.4	50.7	57.1	60.6	59.7	55.6	47.4	39.8	34.2	(33)	
		MCDB	48.1	50.4	57.2	61.6	65.2	73.4	74.2	73.4	70.6	64.9	56.0	47.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	27.4	25.6	27.9	26.5	25.9	30.4	28.1	27.6	28.5	28.5	28.7	26.3	(35)	
		MCDBR	31.8	31.6	33.9	34.3	34.9	36.9	33.4	32.4	33.5	36.0	35.0	33.0	(36)	
	5% WB	MCWBR	17.2	16.2	15.5	14.7	12.9	12.3	10.2	10.3	12.0	15.5	16.8	18.3	(37)	
		MCWBR	31.1	30.8	32.7	31.0	29.0	29.4	26.3	26.0	27.9	30.3	33.6	31.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.230	0.234	0.255	0.278	0.298	0.310	0.337	0.326	0.292	0.260	0.239	0.231	(40)	
		taud	2.635	2.652	2.595	2.542	2.548	2.532	2.501	2.520	2.592	2.661	2.642	2.629	(41)	
	Ebn at Noon		312	324	323	318	311	305	296	298	305	309	306	304	(42)	
		Edh at Noon	21	24	28	31	32	33	33	32	28	23	21	20	(43)	
(44) All-Sky Solar Radiation	RadAvg		862	1170	1558	1866	2060	2304	2166	1967	1708	1284	936	757	(44)	
	RadStd		41	69	115	119	156	107	94	70	88	75	52	40	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (31 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+124	+38	

Nomenclature: See separate page